Kink Instability in Solar Wind Flux Ropes

Code accompanying a paper by Samuel Capellas Coderque, Teresa Nieves-Chinchilla, and Laura Balmaceda, submitted to Solar Physics.

Repo contents

File	Description
`mag_field.py`	Gold-Hoyle magnetic field profiles — Bz, Bθ and their derivatives
`main_notebook.ipynb`	Reproduces all figures in the paper
`realdata_notebook.ipynb`	Applies the analysis to 15 real PSP/SolO ICME observations (Section 4.3). Runs independently and produces Table 1
`icme_instability_results.csv`	Output of `realdata_notebook` — numerical results per event
`icme_tables.tex`	LaTeX table generated from the above, ready to drop into the paper
`sheet_alpha1.csv`, `sheet_alpha1.5.csv`, `sheet_l1e-7.csv`, `sheet_m1.csv`	Pre-computed results used by `main_notebook` to skip slow sections. The code that generates them is included but commented out

Usage

For the figures, run `main_notebook.ipynb` top to bottom. The sheet CSVs are already provided so the slow computations are bypassed automatically.

For Table 1, run `realdata_notebook.ipynb` top to bottom. It produces `icme_instability_results.csv` and `icme_tables.tex`.

Both notebooks import from `mag_field.py` — keep that in the same directory.

Dependencies

A standard scientific Python environment (e.g. Anaconda, Python 3.8+) will have everything needed: `numpy`, `scipy`, `pandas`, `matplotlib`, and `ipywidgets`.